

A certified companion to the Dixmier counterexample in A_3 , and an explicit counterexample to the Poisson conjecture

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Abstract

On 19 July 2026, Alpöge announced an explicit non-injective polynomial map $F: \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 , refuting the Jacobian conjecture in dimension three [2]. The same night, Omniscience Research Agent and Pickhardt [3] carried F across the classical inverse-Jacobian pullback to obtain an explicit injective, non-surjective endomorphism of the third Weyl algebra, refuting the generalised Dixmier conjecture DC_n for $n \geq 3$; priority for that result is theirs. The present note contributes three things. First, a *runnable exact-arithmetic certificate*: a short script that verifies, as polynomial identities over $\mathbb{Q}$, every computational claim on which the construction rests. Second, the *explicit Poisson witness*: applying Bavula’s same-rank associated-graded implication $PC_n \Rightarrow JC_n$ [4, Thm. 7] to Alpöge’s map, we display and certify the resulting injective, non-surjective Poisson endomorphism of the standard symplectic Poisson algebra $\mathbb{C}[x_1, x_2, x_3, \xi_1, \xi_2, \xi_3]$. The transfer mechanism is Bavula’s; as of 21 July 2026 we are unaware of an earlier note displaying and verifying this particular witness. Third, a streamlined self-contained exposition of the transfer. If Zhegllov’s announced proof of DC_1 [9] is confirmed, DC_2 is the only undecided rank.

1 Introduction

Let $A_n = A_n(\mathbb{C})$ denote the n -th Weyl algebra: the unital associative $\mathbb{C}$ -algebra with generators $x_1, \dots, x_n, \partial_1, \dots, \partial_n$ and relations

$$[x_i, x_j] = 0, \quad [\partial_i, \partial_j] = 0, \quad [\partial_i, x_j] = \delta_{ij} \quad (1 \leq i, j \leq n). \quad (1)$$

Dixmier [1] asked in 1968 whether every algebra endomorphism of A_1 is an automorphism; the generalised *Dixmier conjecture* DC_n asserts this for A_n . It is classical that DC_n implies the Jacobian conjecture JC_n in the same dimension (see van den Essen [5]; the construction of the pullback appears in Bavula [4], and the order-preservation argument is recorded by Belov-Kanel and Kontsevich [7, p. 2]), while the much deeper converse passes to doubled dimension: $JC_{2n} \Rightarrow DC_n$ [6, 7]; see also Adjamagbo–van den Essen [8], who add the Poisson conjecture to the equivalence web.

*In collaboration with Claude Fable 5 (Anthropic), which carried out the extraction, the proofs and the symbolic certification. All computations are reproducible from the repository accompanying this note.

On 19 July 2026, Alpöge, working with the language model Claude Fable 5, announced an explicit counterexample to JC_3 [2]: the map $F = (F_1, F_2, F_3): \mathbb{C}^3 \rightarrow \mathbb{C}^3$,

$$\begin{aligned} F_1 &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 z, \end{aligned} \tag{2}$$

satisfies $\det JF \equiv -2$ yet identifies three distinct points:

$$F(0, 0, -\tfrac{1}{4}) = F(1, -\tfrac{3}{2}, \tfrac{13}{2}) = F(-1, \tfrac{3}{2}, \tfrac{13}{2}) = (-\tfrac{1}{4}, 0, 0). \tag{3}$$

That same night, Omniscience Research Agent and Pickhardt [3] published the resulting explicit counterexample to DC_3 , with two complete proofs of non-surjectivity (via centralisers and via the order filtration), a Gröbner-basis verification that the fibre over $(-\frac{1}{4}, 0, 0)$ consists of exactly the three points (3), and the rank monotonicity $DC_n \Rightarrow DC_m$ for $m \leq n$. **Priority for the Dixmier counterexample belongs to [3]**; we learned of their preprint after completing an independent derivation of the same result, and this note has been reframed accordingly.

What this note adds:

1. **A certificate** (Section 2). Reference [3] observes that everything can “be confirmed by exact arithmetic rather than taken on faith” but ships no artefact for doing so. The repository accompanying this note contains a short script certifying, as exact polynomial identities over $\mathbb{Q}$, every computational ingredient: the determinant, the collision, the Weyl relations for the pullback, and the Poisson relations for its graded shadow.
2. **The Poisson instance** (Section 5). The associated-graded map of the Weyl pullback is an explicit injective, non-surjective endomorphism of the polynomial Poisson algebra $\mathbb{C}[x, \xi]$ with its standard symplectic bracket, refuting the Poisson conjecture (as formulated in [8]) in rank ≥ 3 with an explicit witness. The same-rank transfer mechanism — extend the Keller map to the Weyl algebra, pass to the associated graded, observe a Poisson endomorphism — is prior art: it is exactly Bavula’s proof of $PC_n \Rightarrow JC_n$ [4, Thm. 7]. What we contribute is the written-out, certified instance at Alpöge’s map, a consequence not displayed in [3] or, to our knowledge, in the other early consequence notes (e.g. [10], which derives Mathieu-type consequences).
3. **A compact exposition** (Sections 3–4), kept because the certificate refers to its numbered identities.

2 The map, and what is certified

Write $x = (x_1, x_2, x_3) = (x, y, z)$ and $J = JF = (\partial F_i / \partial x_k)_{ik}$. Since $\det J \equiv -2$ (an identity of polynomials; certified, and provable by hand via the elegant substitution of [3, Lem. 1]), the matrix

$$G := (J^\top)^{-1} = -\tfrac{1}{2} \operatorname{adj}(J^\top)$$

has entries in $\mathbb{Q}[x, y, z]$, with $\det G = 1 / \det J^\top = -\tfrac{1}{2}$. The accompanying script `verify_endomorphism.py` certifies, by exact symbolic computation over $\mathbb{Q}$ (every check is a polynomial in $\mathbb{Q}[x, y, z]$ or $\mathbb{Q}[x, \xi]$ tested for literal vanishing):

(C0) $\det J \equiv -2$; the three points in (3) are pairwise distinct and share the stated image;
 $\det G \equiv -\frac{1}{2}$;

(R1) $G J^\top = I$ (nine identities);

(R2) $\sum_k (G_{ik} \partial_k G_{jl} - G_{jk} \partial_k G_{il}) = 0$ for all i, j, l (the flatness identities);

(P1–P3) the Poisson relations of Section 5, checked directly against the definition of the bracket.

The script prints its Python and SymPy versions, exits successfully only if every identity holds, and writes the nine entries of G to a machine-readable file `G.json`, so that the endomorphism below is explicit without any re-derivation by the reader.

3 The Weyl pullback

Define, following the classical construction (Bavula [4]; [3, §3]):

$$\varphi(x_i) = F_i, \quad \varphi(\partial_j) = D_j := \sum_{k=1}^3 G_{jk} \partial_k.$$

Proposition 1. φ extends uniquely to a unital algebra endomorphism of A_3 .

Proof. By the presentation (1) it suffices to check the relations on the images. The F_i commute. Since F_i is a multiplication operator, $[D_j, F_i] = D_j(F_i) = \sum_k G_{jk} \partial F_i / \partial x_k = (G J^\top)_{ji} = \delta_{ji}$ by (R1). For (R2) there is both the certificate and the conceptual argument of [3, Prop. 3]: $[D_i, D_j]$ is a derivation $\sum_l c_l \partial_l$ of $\mathbb{C}[x, y, z]$, and applying it to F_m gives $[D_i, D_j](F_m) = D_i(\delta_{jm}) - D_j(\delta_{im}) = 0$, i.e. $J(c_1, c_2, c_3)^\top = 0$; invertibility of J over $\mathbb{C}[x, y, z]$ forces every $c_l = 0$. $\square$

4 Injectivity and failure of surjectivity

Every element of A_3 has a unique normal form $P = \sum_\alpha a_\alpha(x) \partial^\alpha$; for $P \neq 0$ set $\text{ord}(P) := \max\{|\alpha| : a_\alpha \neq 0\}$ and $\Sigma_m = \{P : \text{ord}(P) \leq m\} \cup \{0\}$. The associated graded algebra is the commutative polynomial ring $\text{gr } A_3 = \mathbb{C}[x_1, x_2, x_3][\xi_1, \xi_2, \xi_3]$ [11, Ch. 2–3]. We identify ξ_j with $-\sigma(\partial_j)$, where σ denotes the principal symbol; with this identification the graded bracket determined by $\{\sigma(P), \sigma(Q)\} = \sigma([P, Q])$ is the standard symplectic bracket of Section 5, with $\{x_i, \xi_j\} = +\delta_{ij}$, and no sign disclaimers are needed below.

Lemma 2. F_1, F_2, F_3 are algebraically independent over $\mathbb{C}$; equivalently $a \mapsto a \circ F$ is injective on $\mathbb{C}[x, y, z]$.

Proof. Standard in characteristic zero from $\det J \neq 0$. A nonzero constant Q cannot satisfy $Q(F) = 0$, so a nonzero relation Q of minimal total degree has positive degree and some $\partial Q / \partial y_i \neq 0$. Applying $\partial / \partial x_k$ to $Q(F) = 0$ gives $\sum_i (\partial Q / \partial y_i)(F) J_{ik} = 0$; invertibility of J over $\mathbb{C}(x, y, z)$ forces $(\partial Q / \partial y_i)(F) = 0$ for all i , contradicting minimality. $\square$

Lemma 3. $\varphi(\Sigma_m) \subseteq \Sigma_m$, the induced graded map is

$$(\text{gr } \varphi)(P)(x, \xi) = P(F(x), G(x) \xi),$$

$\text{gr } \varphi$ is injective, and consequently $\text{ord}(\varphi(P)) = \text{ord}(P)$ for every nonzero $P \in A_3$.

Proof. $\varphi(x_i) \in \Sigma_0$ and $\varphi(\partial_j) \in \Sigma_1$, so φ is filtered and $\text{gr } \varphi$ is the displayed map on $\text{gr } A_3 = \mathbb{C}[x][\xi]$. Suppose $P(F, G\xi) = 0$. Over $K = \mathbb{C}(x, y, z)$ the matrix G is invertible ($\det G = -\frac{1}{2}$), so substituting $\xi \mapsto G^{-1}\xi$ gives $P(F(x), \xi) = 0$ in $K[\xi]$; reading off ξ -coefficients and applying Lemma 2 gives $P = 0$. Strict order preservation follows: a nonzero principal symbol has nonzero image. $\square$

Theorem 4 ([3, Cor. 7]; independent derivation). *φ is injective and not surjective. Hence DC_3 is false, and by padding F with identity coordinates, DC_n is false for every $n \geq 3$.*

Proof. A_3 is simple [1], so the kernel of the unital endomorphism φ is zero. In fact $x_1 \notin \text{im } \varphi$: if $\varphi(P) = x_1$ then Lemma 3 gives $\text{ord}(P) = 0$, so $P = p(x)$ is a polynomial with $p(F) = x_1$. Evaluating at the first two points of (3), whose images under F coincide, gives $p(-\frac{1}{4}, 0, 0) = 0$ and $p(-\frac{1}{4}, 0, 0) = 1$, a contradiction. $\square$

5 The explicit Poisson counterexample

Equip $\mathcal{P}_3 := \mathbb{C}[x_1, x_2, x_3, \xi_1, \xi_2, \xi_3]$ with the standard symplectic Poisson bracket, determined by

$$\{x_i, x_j\} = 0, \quad \{\xi_i, \xi_j\} = 0, \quad \{x_i, \xi_j\} = \delta_{ij}, \quad \{f, g\} = \sum_k \left(\frac{\partial f}{\partial x_k} \frac{\partial g}{\partial \xi_k} - \frac{\partial f}{\partial \xi_k} \frac{\partial g}{\partial x_k} \right).$$

The *Poisson conjecture* in rank n (as formulated in [8]) asserts that every Poisson endomorphism of $\mathcal{P}_n$ — a $\mathbb{C}$ -algebra endomorphism preserving the bracket — is an automorphism.

Theorem 5. *The assignment*

$$\psi(x_i) = F_i, \quad \psi(\xi_j) = \eta_j := \sum_{k=1}^3 G_{jk} \xi_k$$

extends uniquely to a Poisson endomorphism ψ of $\mathcal{P}_3$ (equivalently, $\psi = \text{gr } \varphi$), and ψ is injective but not surjective. Hence the Poisson conjecture is false in rank 3, and, by padding, in every rank $n \geq 3$.

Proof. Well defined. ψ is a polynomial-algebra endomorphism on the 6 generators; it is Poisson iff the brackets of the images match the brackets of the generators. $\{F_i, F_j\} = 0$ since the F_i do not involve ξ (P1). $\{F_i, \eta_j\} = \sum_k (\partial F_i / \partial x_k) G_{jk} = (GJ^\top)_{ji} = \delta_{ij}$ by (R1) (P2). Finally $\{\eta_i, \eta_j\} = \sum_l (\sum_k G_{jk} \partial_k G_{il} - G_{ik} \partial_k G_{jl}) \xi_l$, whose l -th coefficient is the *negative* of the flatness expression (R2), hence vanishes (P3). All three families are also certified directly against the definition of the bracket by the script.

Injective. ψ is exactly the map $P \mapsto P(F, G\xi)$ of Lemma 3, shown there to be injective.

Not surjective. ψ is graded for the ξ -degree: F_i has ξ -degree 0 and η_j has ξ -degree 1, so ψ maps ξ -degree- m homogeneous elements to ξ -degree- m homogeneous elements. Suppose $\psi(P) = x_1$ and write $P = \sum_m P_m$ with P_m homogeneous of ξ -degree m . Since x_1 has ξ -degree 0, we get $\psi(P_m) = 0$ for every $m > 0$; injectivity gives $P_m = 0$, so $P = p(x)$ is a polynomial with $p(F) = x_1$. Evaluating at the first two points of (3), whose images under F coincide, gives $p(-\frac{1}{4}, 0, 0) = 0$ and $p(-\frac{1}{4}, 0, 0) = 1$, a contradiction. Hence $x_1 \notin \text{im } \psi$.

For $n \geq 4$, pad F with identity coordinates as in Theorem 4; the same argument applies verbatim. $\square$

Remark 6. The transfer mechanism in Theorem 5 is not new: extending a Keller map to the Weyl algebra, passing to the associated graded, and observing a Poisson endomorphism whose invertibility would force invertibility of the original map is precisely Bavula’s same-rank proof of $PC_n \Rightarrow JC_n$ [4, Thm. 7]; no stable equivalence and no rank doubling enters. Our contribution is the explicit certified instance at the map (2). With the identification of Section 4, $\psi = \text{gr } \varphi$ literally: the counterexample to the quantum statement and the counterexample to its classical limit are associated graded to one another.

Remark 7 (Status of the remaining ranks). Rank monotonicity ($DC_n \Rightarrow DC_m$ for $m \leq n$, [3, Prop. 9]) means the refutation propagates upward only. Zhiglov has posted a preprint claiming a proof of DC_1 [9]; we do not assess its status. If that claim is confirmed, DC_2 is the only undecided rank of the Dixmier family, as JC_2 is of the Jacobian family.

Code availability

`verify_endomorphism.py` and `G.json` accompany this note. A kernel-checked (Lean 4) rendering of the certified identities is mechanical and planned as a follow-up; we make no claim for it here.

Acknowledgements

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References

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